

Capstone Project Plan
Ski Safety GPS

Group 6:

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Professor Mark Thomas

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Abstract

The proposed capstone project focuses on developing an innovative ski safety device that combines relative affordability, reliability, and advanced features specifically tailored to the skiing and intense winter sports communities. The envisioned product is a safety device that utilizes GPS and LoRaWAN technologies to provide location tracking, automatic emergency detection, and manual alert capabilities all in a wristband form factor. Our aim is to address existing gaps in the skiing market and deliver a solution that is both cost effective and functionally superior to current inventions.

Market Analysis

Existing products, such as Garmin's handheld GPS devices and SPOT satellite communications, serve outdoor enthusiasts well but with expensive and overly complex solutions. Other devices such as the skiKrumb, cater to real-time location tracking for individual families and instructors but lack features such as emergency detection and affordability.

Insights

- **Cost:** Most GPS products exceed affordable price points, placing them out of reach for budget conscious consumers.
- **Emergency Detection:** Very few of these devices on the market include automatic emergency detection systems.
- **Accessibility:** Current products often rely on bulky designs and lack ease of access in emergency alert situations.

Proposed Solutions

- **Affordability:** Utilizing the low-cost RAK3172 LoRaWAN module, we aim to keep the price per unit below \$100 CAD.
- **Automatic Emergency Detection:** Incorporating sensors to detect crashes or prolonged inactivity, the device will automatically alert nearby safety instructors or staff members.
- **Accessibility:** The compact wristband form factor ensures ease of use, while an accessible panic button provides reassurance for manual emergency situations.
- **Durability:** Designed for even the most extreme skiing condition, the device will feature adequate waterproofing, impact resistance, and an operational temperature range of -40°C to +85°C

Technology Integrations

To smoothly further development, we will leverage on existing open-source materials, such as the [Loko](#) tracker project on GitHub. This will allow us to focus on enhancing the core functionalities while reducing development time and costs. The RAK3172 LoRaWAN

module will enable long-range, low-power communications, ensuring that there is reliable performance in relatively remote locations. Furthermore, the compact design of the module supports portability and aligns with our end goal of creating a wristband device.

Niche Market

Our primary audience includes but is not limited to expert class skiers and snowboarders seeking additional security measures, budget-conscious recreational users, and ski resorts aiming to enhance their safety protocols. By addressing limitations in the current market such as cost, ease of use, and durability, our product fills a unique gap and avoids direct competition with well-established brands offering a more general solution.

This project aims to deliver a unique, niche focused ski safety device that meets the specific needs for the skiing community. By addressing cost, ease of use, and functionality gaps in existing offerings, our solution has the potential to break through in snow sports, providing peace of mind and enhancing security for every user.

Approval and suggestions

This project represents the culmination of extensive research, drawing from prior similar initiatives and integrating valuable insights from Mike. Leveraging these insights, we proposed a novel concept, which was subsequently approved and further refined by Professor Mark Thomas. Upon reviewing the initial draft of the project, suggestions were given for potential technical enhancements, particularly focusing on optimizing cost-effectiveness.

A primary area of improvement identified was the LoRaWAN modules, as they can be quite expensive. To put things into perspective, the average size of a ski resort ranges from 100 to 1,500 acres, which translates to approximately 0.4 to 6.07 square kilometers. The RAK3172 module, with its impressive range of up to 10 kilometers, is more than capable of covering the entirety of most ski resorts. In cases where specific challenges arise, the addition of more gateway units can ensure seamless functionality across larger or more complex terrains.

Moreover, one of the standout features of the RAK3172 is its remarkably low power consumption. This characteristic reduces the frequency of recharging, making it a practical and efficient choice for this application.

Professor Mark and other advisors also provided a critical suggestion that broadened the scope of our project. They emphasized that while ski resorts serve as an excellent initial use case, the project need not be limited to this domain. With a few adjustments and customizations, the system can be adapted to suit a variety of extreme sports and outdoor activities. This expanded versatility opens up numerous opportunities for application in other challenging environments, enhancing the potential impact and market reach of our solution.

Scope Management Plan

Requirements Collection

The goal is to systematically gather comprehensive requirements for the ski safety wristband device by identifying, capturing, and validating all essential requirements. This can ensure that both the final product and the process fully meets the users end needs and align with stakeholder expectation. Our approach will focus on:

Key Requirement Gathering Methods:

- Direct interviews with ski resort safety managers
- Comprehensive surveys of skiers (both expert and recreational)
- Detailed market analysis of existing safety devices
- Technical consultations with winter sports equipment experts
- Distribute online and on-site surveys to skiers to gather quantitative feedback on features
- Review existing winter sports safety products and wearable devices to benchmark features

Critical Requirements:

- Reliable GPS tracking functionality
- Automatic emergency detection system
- Manual alert mechanism
- Cost-effective production (under \$100 CAD)
- Compact wristband design
- Extreme weather durability

Scope Definition

The project aims to design, develop, and deliver a wearable ski safety wristband that enhances skiers' safety by detecting crashes, pinpointing locations, and providing timely alerts to ski resort safety persons. By integrating LoRaWAN communication and GPS tracking in a robust, user-friendly device, this project will address the business need to improve skier safety while reducing the reliance for on-site staff:

Business Objectives:

- Improve Ski Slope Safety: Decrease the response times to accidents and enhance overall skier safety standards.
- Reduce Operation Costs: Lower labor costs by minimizing the need for additional on-site staff.
- Enhance Customer Satisfaction: Build a trustable device that offers reliable safety measures and reassures the user of prompt assistance in cases of emergency.

- Completed Project: Launch a functional prototype by the end of the project due date.

Project Deliverables:

- Wearable Wristband: A fully functional, weather-resistant, and compact device integrating LoRaWAN module
- Crash/Inactivity Detection: Embedded software capable of detecting collisions and extended inactivity
- Manual Alert: A user-friendly interface that sends distress signals

Project Exclusions:

- Full smartphone application
- Advanced fitness tracking
- Global satellite communication
- Comprehensive medical response systems

Constraints:

- Time and Schedules: Must adhere to the planned timeline with project completion by the course due date
- Cost: Each device must be produced at or under \$100 CAD to maintain cost-effective pricing
- Environmental Conditioning: Device must function reliably at extreme temperatures
- Scope and Resources: Limited engineering and development resources.

Assumptions:

- Reliable LoRaWAN Infrastructure: Ski resorts have or will implement sufficient LoRaWAN gateways to enable communication with the wristband
- User Adoption and Feedback: Users and resort staff will provide timely feedback for project refinement of features and performance
- Industry Standards: All relevant safety, electronics, and communications standards can be met.

In-Scope Components:

1. Hardware and Core Function

- Wristband device development: Design and prototype a compact, weather-resistant wearable device that is easily attachable to the user's wrist.
- LoRaWAN technology integration: Utilize LoRaWAN modules and gateways for low-power, long-range communication between the wristband and the end device monitoring computer.
- Crash and inactivity detection algorithms: Develop accelerometer-based program to recognize sudden impacts or extended motionlessness, triggering the automatic

alert. Fine tune false alerts where most alerts are certain.

- Emergency communication protocols: Install a button that allows skiers to manually send a distress signal.

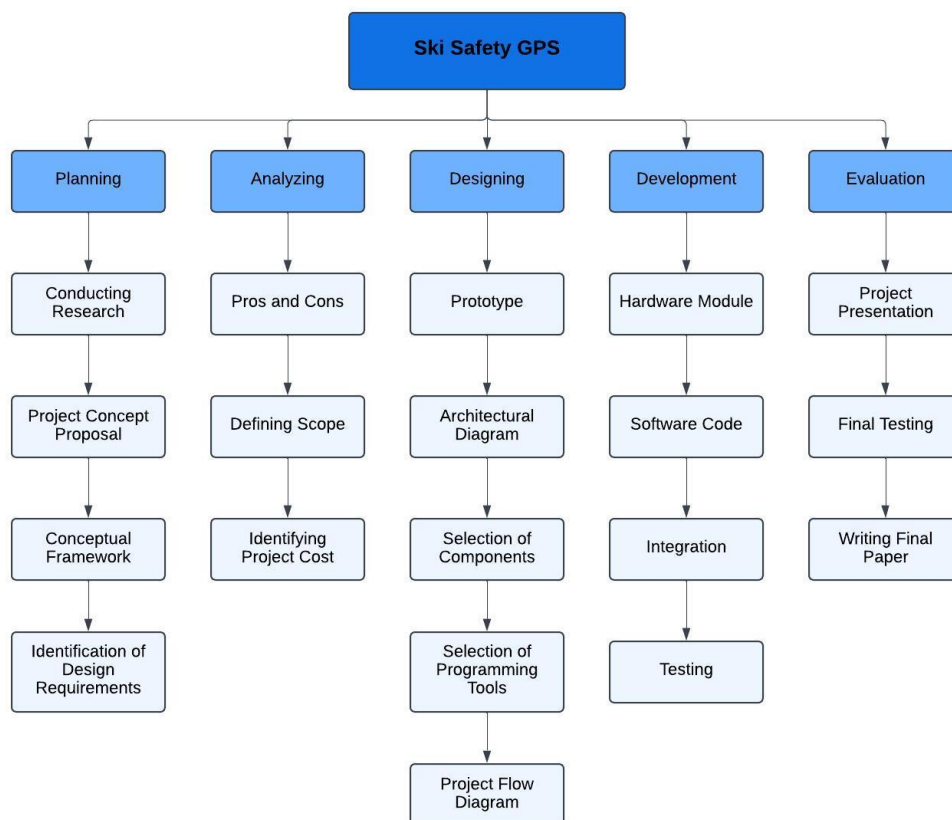
2. Durability

- Robust physical design (water-resistant, impact-resistant): Resistant up to IP56 rating.
- Operational in extreme temperatures (-40°C to +85°C)

Out of Scope:

- Full smartphone application
- Advanced fitness tracking
- Global satellite communication
- Comprehensive medical response systems

WBS



Scope Control

To ensure that the project's defined scope remains stable and aligns with the business objectives, preventing unwanted expansion or the inclusion of features that are out of the constraints:

Control Mechanisms:

- Establish formal change management process: Each change must be submitted into a formal change request process that is documented and reviewed to properly match current scope standards.
- Create detailed project charter
- Define clear decision-making protocols
- Maintain regular stakeholder communication: Monthly meetings including the Project Manager, and technical leads to converse with stakeholders to review purposed changes.
- Conduct quarterly technical and market reviews

Control Checkpoints:

- Budget monitoring: Monthly budget reviews to ensure that all scope activities are within the financial constraints.
- Development milestone tracking: Assess each major milestone to confirm the deliverables that meet the defined scope criteria.
- User feedback integration: Evaluate end-user and stakeholder feedback using an online survey.
- Market relevance assessment: Review the market competitors and confirm whether the current scope remains relevant and valuable to the skier space. Propose any scope changes in response to shifting markets.

Governance Structure:

- Monthly project review meetings: Ensures that any scope requirements that need to be adjusted are addressed.
- Documented change request procedures: All scope related decisions and outcomes are documented in the project repository.
- Clear escalation paths: If concerns are caused by scope changes, defined escalation procedures are taken place to timely resolve the issue at appropriate management levels.
- Transparent communication channels: Stakeholders receive regular updates on the scope stability, approved changes, and reasons behind accepted and rejected modifications.

The primary objective remains developing an innovative, affordable ski safety device that addresses critical market gaps in winter sports safety technology.

Scope Validation

Ensuring the product meets market needs and technical specifications. Validation ensures that the wristband's performance matches the established scope, safety standards, and user

expectations:

Validation Approaches:

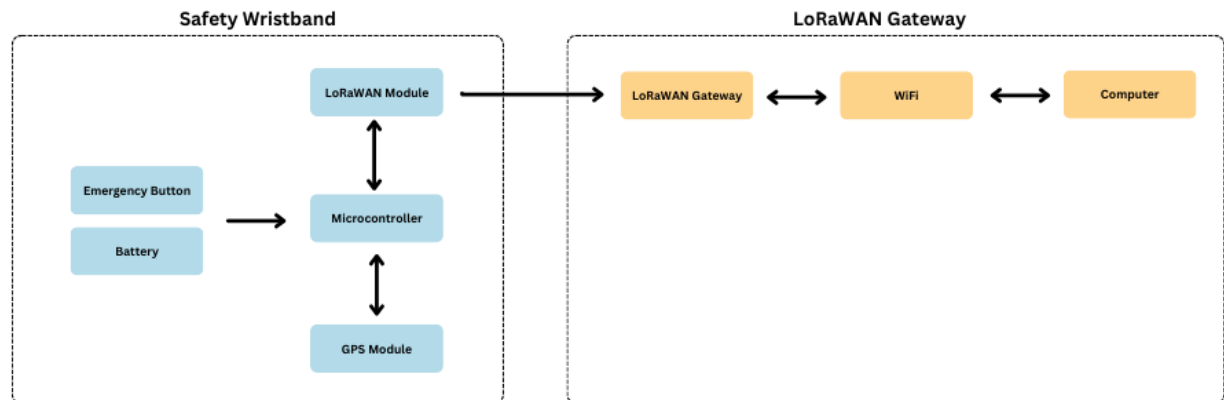
- Prototype testing with experienced skiers (At least 2-5 years of experience): Equip 3-4 skiers with a prototype device for 3-5 hours under various conditions.
- User experience focus groups: Conduct group sessions with at least 5 participants, measuring average time of response to manual alert presses.
- Technical feasibility assessments
- Cost-effectiveness analysis
- Safety regulation compliance checks

Validation Criteria:

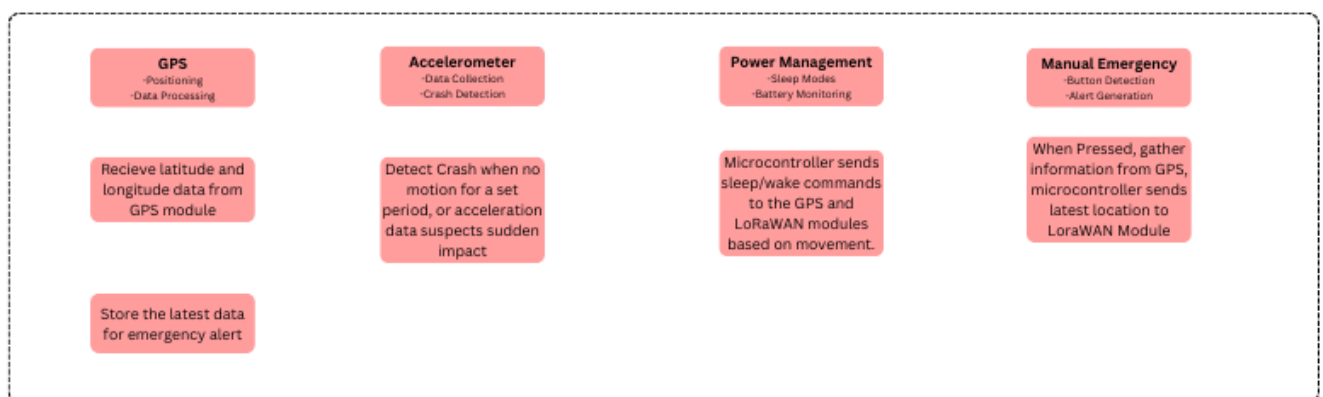
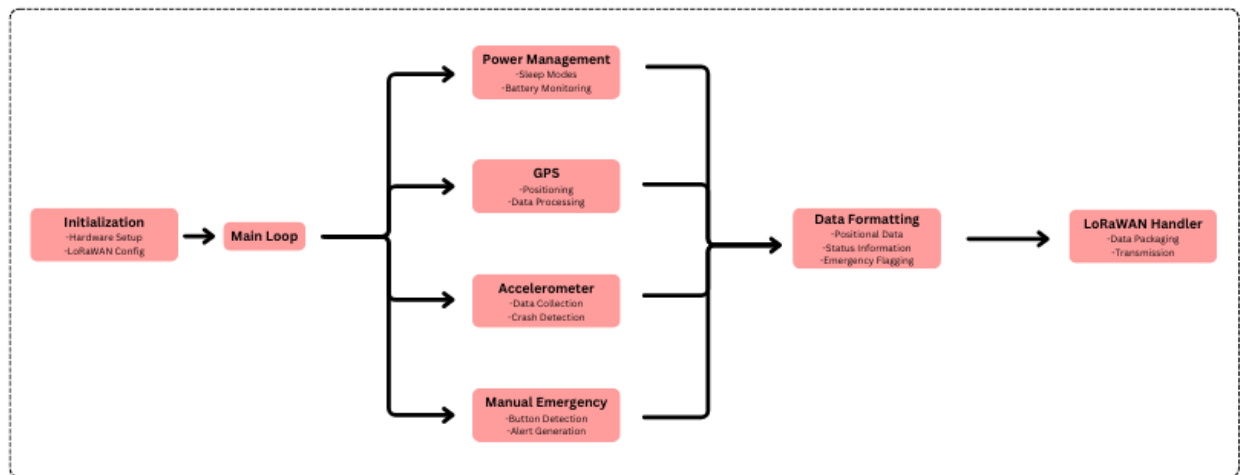
- Emergency detection accuracy: >95% of crash simulations correctly trigger alerts within less than 5 seconds proceeding crash.
- GPS Precision: Device must report user's location with +/- 10 meters of real position 90% of the time.
- Latency: Data packets must reach the monitoring computer within 10 seconds.
- Performance in extreme conditions:
 - Verify that each device continuously operates at -40°C for at least 3 hours.
 - Withstand 5 drops from 1.5 meters without losing main functionality (e.g., manual button, location tracking, crash detection)
- Interface Performance: Manual button presses must be detected and send a signal to the main computer 100% of the time.
- Cost target achievement: Total production cost per unit <\$100 CAD
- Alignment with ski resort safety standards: All electronic components must align with the ski resort standards.

Block Diagram

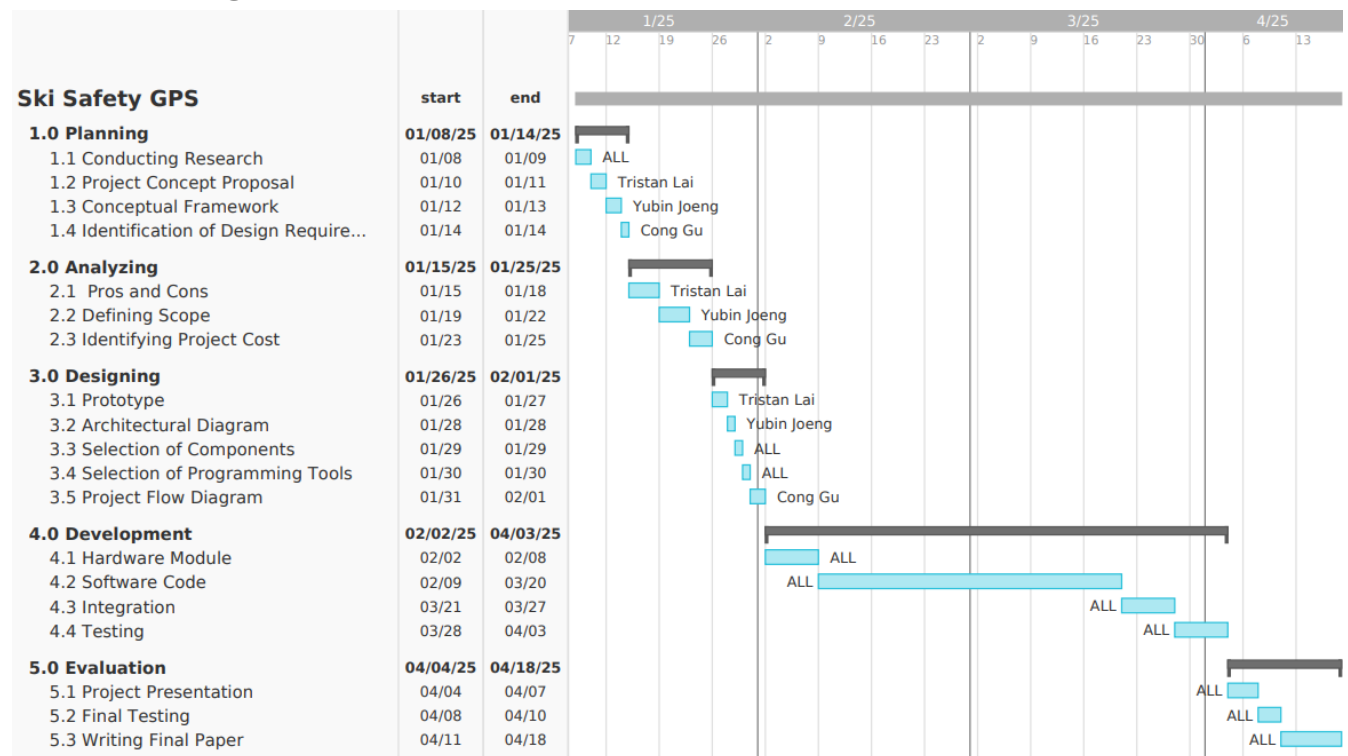
Hardware Block Diagram



Software Block Diagram



Time Management Plan



1.0 Planning

1.1

- All members gather background information on GPS modules, sensors, and other components.
- All members gather information on potential competitors.

1.2

- Tristan drafts a project concept proposal, highlighting key deliverables and encapsulating vital components of the project.

1.3

- Yubin defines the technical framework, identifying the required hardware and software that is required.

1.4

- Cong compiles all research and technical requirements to start a project scope and finer details.

2.0 Analyzing

2.1

- Tristan researches into current competitors with their pros and cons. Finding alternative features that will separate the project from current offerings.

2.2

- Yubin documents scope boundaries and deliverables that align with the small form factor

and robust features.

2.3

- Cong estimates each component costs and the labour required to implement them
- Documenting the business and technical priorities.

3.0 Designing

3.1

- Tristan will create the initial breadboard prototype.
- Test any inconsistencies in the project and document.

3.2

- Yubin will design block diagrams showing data flow between each sensor, microcontroller, and output to the LED.

3.3

- All members are responsible for research and finalization of the selection of GPS module, microcontroller, battery and other components.

3.4

- All members will research the proper programming tools and libraries necessary for programming the Ski Safety GPS.
- Finding existing GitHub projects that may be beneficial for the project.

3.5

- Cong will create a detailed flowchart representing system behaviours, decision points and other important processes.

4.0 Development

4.1

- All members will assemble the PCB, solder components, conduct basic continuity tests.

4.2

- Develop core firmware (Yubin)
- Write the distress signal logic (Tristan)
- Handle communication between the device and server (Cong)

4.3

- All members will combine hardware and software components, ensuring data function.

4.4

- All members will conduct simple tests for the developed device. Testing for performance and reliability, ensuring all deliverables are met.

5.0 Evaluation

5.1

- All members will develop slides covering project objectives, technical details, and test results.
- Prepare a script for the presentation.

5.2

- All members will perform field tests in a ski-like environment, documenting the performance and overall reliability.

5.3

- All members will compile all project documentation, summarizing the findings, challenges and future improvements that could be made.

PERT

Optimistic (O): The shortest possible duration

Most Likely (M): The best estimate of the duration

Pessimistic (P): The longest duration if significant issues arise.

Expect Time (TE): $TE = O + 4M + P / 6$

Phase	Description	O (days)	M (days)	P (days)	TE Calculation	TE (days)
Planning	Research, concept proposal, and requirement identification	5	7	10	$(5 + 4*7 + 10)/6 = 43/6$	7.17
Analyzing	Finalizing requirements, defining scope, identifying project costs	9	11	15	$(9 + 4*11 + 15)/6 = 68/6$	11.33
Designing	Developing prototype, component selection, detailed diagrams	17	21	30	$(17 + 4*21 + 30)/6 = 131/6$	21.83
Development	Assembly, firmware coding, integration	50	61	75	$(50 + 4*61 + 75)/6 = 369/6$	61.5
Evaluation	Field Testing, validation against specs, final documentation	10	15	20	$(10 + 4*15 + 20)/6 = 90/6$	15

Cost Management

Roles and Responsibilities for Cost Management

- Project Manager (PM): Oversees the cost management process approves budget changes, manages contingencies and reserves.
- Electronics Engineer (EE): Provides material cost estimates, identifies technical risk areas that may lead to cost overruns.
- Software Developer (SD): Estimates and tracks software development hours, estimates delays that might increase labour costs

Cost Estimation

Labor Rates (Estimated)

- Project Manager: \$25/hr
- Electronics Engineer: \$20/hr
- Software Developer: \$20/hr

Work Hours

Assuming the project runs for approximately 14 weeks and each member contributes 10 hours/week .

PM: 140 hours * \$25/hr = \$3,500

EE: 140 hours * \$20/hr = \$2,800

SD: 140 hours * \$20/hr = \$2,800

Total Estimated Labour Costs: \$9,100

Unique Number	Description	Requirement Origin	Business Priority	Technical Priority	Cost	Phase, Release, Iteration
S1	Emergency Button	Safety	High	Low	\$5	Prototyping
C1	LoRaWAN Module	Connectivity	High	High	\$6	Prototyping
C2	LoRaWAN Gateway	Connectivity	High	High	\$50	Prototyping
H1	Designed Board	Hardware	High	High	\$20	Prototyping
U1	Wrist Band	User Experience	Low	Low	\$10	Assembly, Testing
C3	Antenna	Connectivity	Medium	Medium	\$5	Prototyping
U2	Case	User Experience	Medium	Low	\$8	Assembly, Testing
U3	Battery Life Monitoring	User Experience	Low	Low	\$0	Prototyping
U4	Push Notifications	User Experience	Medium	Medium	\$0	Prototyping
S2	Collision Detection Software	Safety	High	High	\$0	Prototyping
P1	Arduino	Prototyping	High	High	\$20	Prototyping
U5	System Firmware	User Experience	High	Medium	\$0	Prototyping

TOTAL AMOUNT (CAD)	\$124 (±10%)
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Additional Costs

Testing and Field Trials: Renting ski slope time, protective gear for testers. Est. \$500

Software/Tools: Assumed \$0; provided by Centennial College

Risk Contingencies

- Hardware: Risk of component failure or design reworks. 10% contingency of hardware and material costs
- Environmental Testing: If the prototype fails environmental stress tests additional costs for components and re-tests are required
- 10% of Labour = $\$9,100 * 0.10 = \910
- 10% of Materials = $\$124 * 0.10 = \12
- 10% of Testing = $\$500 * 0.10 = \50

Total Contingency = \$910 (labor) + \$12 (materials) + \$50 (testing) = \$972

Updated Subtotal: \$9,724 (Direct) + \$972 (Contingency) = \$10,696

Cost Control

Earned Value Management

- Track Planned Value, Earned Value, and Actual Costs monthly
- Calculate the Cost Performance Index and Schedule Performance Index

Control Process

- Any changes that impact on the cost must be requested via a Change Request Form.
- The PM and stakeholders must evaluate the cost impact
- Monthly Cost Reviews: Compare actual spending vs. baseline.

Note: The labour and testing costs presented here are hypothetical estimates, included only to demonstrate how costs might be represented and managed in a real-world scenario.